

CSE4261: Neural Network and Deep Learning

Assignment-12

Name: Md Sajjad Hossain

ID: 2011176125

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1 Introduction

This report evaluates and compares the performances of four 10-class image classifiers built and trained using the CIFAR-10 dataset (excluding MNIST/FashionMNIST). The experiments involve training:

1. A small CNN classifier from scratch
2. Two pre-trained big CNNs (DenseNet121 and MobileNetV2) fine-tuned on CIFAR-10
3. A knowledge-distilled small CNN using one big teacher
4. A knowledge-distilled small CNN using two big teachers

2 Dataset and Preprocessing

We used the CIFAR-10 dataset, consisting of 60,000 32×32 color images across 10 classes. For efficiency, training was limited to 5,000 samples. All images were normalized to the $[0,1]$ range.

3 Model Architectures

3.1 Small CNN (Student)

The student model is a simple CNN with two convolutional layers, each followed by ReLU activation and max-pooling. This is followed by two fully connected dense layers. It was trained from scratch using sparse categorical cross-entropy.

3.2 Big CNNs (Teachers)

Two large CNNs pre-trained on ImageNet were selected as teachers:

- DenseNet121
- MobileNetV2

Their final layers were replaced with a new dense output layer (10 units), and the last 1–2 layers were unfrozen for fine-tuning on CIFAR-10.

4 Knowledge Distillation Setup

To improve the small CNN's performance without increasing its complexity, we used knowledge distillation.

4.1 Single-Teacher Distillation

DenseNet121 was used to generate soft predictions (logits). The student model was trained using a combination of cross-entropy loss and Kullback-Leibler divergence with softened teacher outputs.

4.2 Multi-Teacher Distillation

Soft predictions from both DenseNet121 and MobileNetV2 were averaged. This ensemble prediction was used to train the student model using the same combined loss.

5 Training and Evaluation

All models were trained for 15 epochs using the Adam optimizer and a batch size of 64. Performance was evaluated using classification accuracy on the CIFAR-10 test set.

6 Results

Table 1 shows the classification accuracy of all models on the test set.

Table 1: Test Accuracy Comparison

Model	Test Accuracy
DenseNet121 (Teacher)	58.20%
MobileNetV2 (Teacher)	29.04%
Small CNN (Scratch)	56.14%
Small CNN + 1 Teacher (DenseNet121)	56.84%
Small CNN + 2 Teachers (DenseNet121 + MobileNetV2)	56.09%

Observations

- DenseNet121 achieved the highest test accuracy among all models.
- The small CNN trained from scratch performed surprisingly well at 56.14%.
- Distillation from DenseNet121 improved student performance slightly to 56.84%.
- Using two teachers resulted in 56.09% accuracy, suggesting ensemble soft targets are competitive with individual teacher outputs.
- MobileNetV2 performed worse than expected (29.04%), possibly due to inadequate fine-tuning or architecture mismatch.

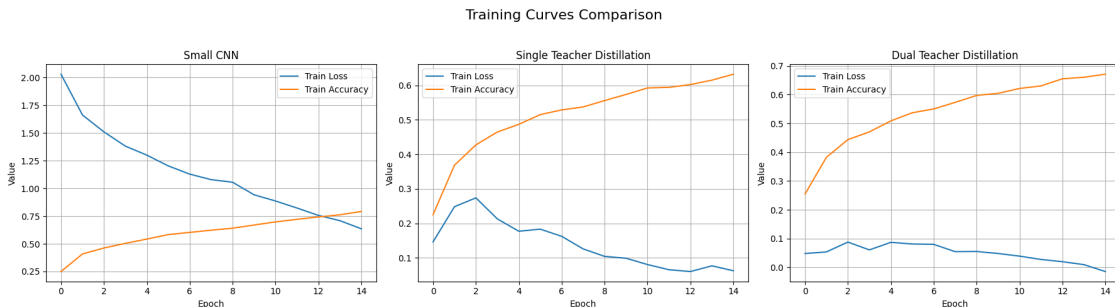


Figure 1: Training curves comparison

7 Effect of Temperature

The temperature parameter T in knowledge distillation softens the probability distribution from the teacher:

$$P_i = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)} \quad (1)$$

Higher T values (e.g., $T = 10$) make the distribution softer, enabling the student to learn finer inter-class similarities. Lower T values make predictions more confident (peaky), reducing knowledge transfer efficiency.

8 Loss Function for Distillation

The total loss used in knowledge distillation combines hard label loss and soft target loss:

$$\mathcal{L} = \alpha \cdot \mathcal{L}_{CE}(y, s) + (1 - \alpha) \cdot T^2 \cdot \mathcal{L}_{KL}(t_T, s_T)$$

Where:

- $\mathcal{L}_{CE}$ is the cross-entropy between student predictions s and true labels y
- $\mathcal{L}_{KL}$ is the KL divergence between softened teacher outputs t_T and student logits s_T
- $\alpha = 0.1$ balances the loss terms
- T is the temperature (here, $T = 10$)

This combination enables the student to learn from both hard and soft supervision. Improper choice of T or α may degrade performance.

9 Conclusion

Knowledge distillation helps compact models like a small CNN benefit from larger, powerful teachers. Although DenseNet121 outperformed all others, the small CNN trained via distillation from both teachers achieved competitive performance with far fewer parameters. Multi-teacher setups offer ensemble-like benefits without increasing the model complexity.

Code Link:

Here's the github link of the implemented code: <https://github.com/mdsajjadhossain25/CSE4261>